DO NOW, AO1 REVISION

Half-lives and the random nature of radioactive decay

Atomic structure
Paper 1, Combined and Separate
8 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 2 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**Define half-life in terms of activity.**	The time taken for the activity (or count rate) of a sample to halve.	☐
2 ★	**Define half-life in terms of the number of nuclei.**	The time taken for the number of nuclei of the isotope in a sample to halve.	☐
3	**Describe what happens to the half-life of an isotope as time passes.**	Nothing: it is constant.	☐
4	**Describe what happens to the activity of a radioactive sample over time.**	It decreases.	☐
5	**Describe what happens to the number of unstable nuclei in a radioactive sample over time.**	It decreases.	☐
6	**State the fraction of radioactive nuclei remaining after two half-lives.**	One quarter (¼)	☐
7	**State the fraction of radioactive nuclei remaining after three half-lives.**	One eighth (⅛)	☐
8 ★	**Describe how to find the half-life from a graph of activity against time.**	Find the time taken for the activity to fall to half its starting value.	☐

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